

Completion of the proof of ??: ramification after blowup is trivial for $G = \mathbb{Z}/p^r\mathbb{Z}$

a direct proof via Artin–Schreier–Witt theory

July 6, 2026

Goal. Let C be a projective smooth geometrically connected curve over k ($\text{char } k = p$), $\mathfrak{m} = \sum n_i P_i$ a modulus, $U = C \setminus \mathfrak{m}$, and let $\mathcal{P} \rightarrow U$ be a G -torsor with ramification bounded by $\mathfrak{m}$, where $G = \mathbb{Z}/p^r\mathbb{Z}$. Fix $P = P_i$, put $n = n_i$, $\mathfrak{n} = nP$ and $d = \deg \mathfrak{n} = n$. Let $X_{\mathfrak{n}} \rightarrow C^{(d)}$ be the blowup at the point $\mathfrak{n} \in C^{(d)}$, with exceptional divisor $E_{\mathfrak{n}}$ and generic point $\eta_{\mathfrak{n}}$. We prove:

Theorem (= ??). $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$.

Strategy. The thesis already proves this for $r = 1$ by an explicit computation: the Artin–Schreier datum of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ is the power sum $\sum_{i=1}^d f(x_i)$, which is a symmetric function of low degree in the $y_i = 1/x_i$ and therefore lands in the *residue field* of the exceptional valuation. The inductive route from $\mathbb{Z}/p$ to $\mathbb{Z}/p^r$ stalls (see `a_wrong_path.tex`); instead we redo the $r = 1$ computation verbatim with a length- r *Witt vector* in place of a single Artin–Schreier equation. The datum of $\mathcal{P}^{[d]}$ becomes the d -fold *Witt sum* of the local datum of $\mathcal{P}$; the Witt carries are controlled by the isobaric grading of the Witt addition polynomials, and the ramification bound $< d$ makes every component of the datum a symmetric polynomial of degree $< d$ in the y_i — hence integral at $\eta_{\mathfrak{n}}$, hence unramified, because the Artin–Schreier–Witt isogeny is étale.

No Swan conductors are used. The only imported result is the classical Schmid–Witt computation of ramification breaks of Artin–Schreier–Witt extensions ([Tho05]), whose $r = 1$ case is ?? of the thesis. Sections 1 and 2 are supplementary background (intended for the Preliminaries); Sections 3 and 4 contain the proof.

1 Background I: Witt vectors

Fix a prime p . We recall the p -typical Witt vectors of finite length $r \geq 1$; the standard reference is [Ser79, II §6] (see also [Rab14]).

For a commutative ring A , the set $W_r(A) = A^r$ of vectors $\underline{a} = (a_0, \dots, a_{r-1})$ carries a commutative ring structure, functorial in A , uniquely determined by the requirement that the *ghost maps*

$$w_n: W_r(A) \longrightarrow A, \quad w_n(\underline{a}) = \sum_{l=0}^n p^l a_l^{p^{n-l}} = a_0^{p^n} + p a_1^{p^{n-1}} + \dots + p^n a_n \quad (0 \leq n \leq r-1)$$

are ring homomorphisms. Concretely, there are universal polynomials with *integer* coefficients computing the ring operations. We will only need addition and subtraction, and only their following structural features.

Notation. For $d \geq 2$ let $S_n^{(d)} \in \mathbb{Z}[X_l^{(i)} : 1 \leq i \leq d, 0 \leq l \leq r-1]$, $0 \leq n \leq r-1$, denote the d -fold Witt *addition* polynomials, characterized by

$$w_n(S_0^{(d)}, \dots, S_n^{(d)}) = \sum_{i=1}^d w_n(\underline{X}^{(i)}), \quad (1)$$

so that $\underline{X}^{(1)} +_W \dots +_W \underline{X}^{(d)} = (S_0^{(d)}, \dots, S_{r-1}^{(d)})$. Similarly let $D_n(\underline{X}; \underline{Y})$ denote the subtraction polynomials, characterized by $w_n(\underline{D}) = w_n(\underline{X}) - w_n(\underline{Y})$. We write $+_W$, $-_W$ and $\sum^W$ for Witt-vector operations, to distinguish them from componentwise ones, and $\underline{0} = (0, \dots, 0)$.

Expanding (1) and using $w_n(\underline{X}) = p^n X_n + \sum_{l < n} p^l X_l^{p^{n-l}}$ we get the recursion (valid in $\mathbb{Q}[X_l^{(i)}]$, where the polynomials are uniquely determined):

$$p^n S_n^{(d)} = \sum_{i=1}^d w_n(\underline{X}^{(i)}) - \sum_{l=0}^{n-1} p^l (S_l^{(d)})^{p^{n-l}}, \quad (2)$$

and the analogous recursion for D_n with $\sum_i w_n(\underline{X}^{(i)})$ replaced by $w_n(\underline{X}) - w_n(\underline{Y})$. That these recursions produce polynomials with integer coefficients is Witt's theorem [Ser79, II §6, Thm. 6].

Definition 1 (weights). Assign to the level- l variable $X_l^{(i)}$ (resp. X_l, Y_l) the *weight* p^l . The weight of a monomial is the sum of the weights of its variables, counted with multiplicity. A polynomial is *isobaric of weight* w if every monomial occurring in it has weight exactly w . (In particular an isobaric polynomial of weight $w > 0$ has no constant term.)

Lemma 2 (isobaricity). $S_n^{(d)}$ and D_n are isobaric of weight p^n .

Proof. The ghost polynomial w_n is isobaric of weight p^n : its monomial $p^l X_l^{p^{n-l}}$ has weight $p^l \cdot p^{n-l} = p^n$. We induct on n using (2). For $n = 0$, $S_0^{(d)} = \sum_i X_0^{(i)}$ is isobaric of weight 1. For $n > 0$: each $w_n(\underline{X}^{(i)})$ is isobaric of weight p^n ; by induction $S_l^{(d)}$ is isobaric of weight p^l for $l < n$, hence $(S_l^{(d)})^{p^{n-l}}$ is isobaric of weight $p^l \cdot p^{n-l} = p^n$. So the right-hand side of (2) is isobaric of weight p^n in $\mathbb{Q}[X_l^{(i)}]$, and dividing by p^n does not change the set of monomials. The same induction applies to D_n . $\square$

Lemma 3 (triangularity). 1. $S_n^{(d)} = \sum_{i=1}^d X_n^{(i)} + T_n$ with $T_n \in \mathbb{Z}[X_l^{(i)} : l < n]$, and $D_n = X_n - Y_n + E_n$ with $E_n \in \mathbb{Z}[X_l, Y_l : l < n]$. In words: the n -th component of a Witt sum or difference is the plain sum or difference of the n -th components, plus an integer polynomial in the components of levels $< n$.

2. Consequently, if $\underline{u}, \underline{v} \in W_r(A)$ satisfy $u_l = v_l = 0$ for all $l < n$, then $\underline{u} \pm_W \underline{v}$ agrees with the componentwise sum or difference up to and including level n .

Proof. (1) Induct on n via (2): in $w_n(\underline{X}^{(i)}) = p^n X_n^{(i)} + (\text{polynomial in } X_l^{(i)}, l < n)$ only the displayed term involves level n , and the correction terms $(S_l^{(d)})^{p^{n-l}}$, $l < n$, involve only levels $\leq l < n$ by induction. Divide by p^n ; integrality of the result is part of Witt's theorem. (2) By (1), the components of level $\leq n$ of $\underline{u} \pm_W \underline{v}$ are $u_l \pm v_l + (\text{polynomial without constant term in levels } < l)$, and the correction vanishes when all lower components vanish (Lemma 2 guarantees there is no constant term). $\square$

Frobenius and the operator $\wp$

From now on A is an $\mathbb{F}_p$ -algebra. The absolute Frobenius $\phi: A \rightarrow A$, $a \mapsto a^p$, is a ring endomorphism, so by functoriality it induces a ring endomorphism

$$F := W_r(\phi): W_r(A) \rightarrow W_r(A), \quad F(a_0, \dots, a_{r-1}) = (a_0^p, \dots, a_{r-1}^p),$$

which acts *componentwise*. (For $\mathbb{F}_p$ -algebras this componentwise map coincides with the Witt-vector Frobenius of the general theory [Ser79, II §6]; we will only ever use the componentwise description, which we may take as the definition.) Define the *Artin–Schreier–Witt operator*

$$\wp := F - \text{id}: W_r(A) \longrightarrow W_r(A), \quad \wp(\underline{a}) = F(\underline{a}) - \underline{a}.$$

Since F is a ring homomorphism, $\wp$ is an endomorphism of the additive group $W_r(A)$, functorial in A . For $r = 1$ it is the classical Artin–Schreier operator $a \mapsto a^p - a$.

Lemma 4 (shape of $\wp$). *For variables $\underline{Y} = (Y_0, \dots, Y_{r-1})$ over $\mathbb{F}_p$,*

$$\wp(\underline{Y})_n = Y_n^p - Y_n + \Delta_n(Y_0, \dots, Y_{n-1}), \quad 0 \leq n \leq r-1,$$

where $\Delta_n \in \mathbb{F}_p[Y_0, \dots, Y_{n-1}]$ has no constant term and $\Delta_0 = 0$.

Proof. $\wp(\underline{Y})_n = D_n(Y_0^p, \dots, Y_{r-1}^p; Y_0, \dots, Y_{r-1})$. By Lemma 3(1) this equals $Y_n^p - Y_n + E_n(Y_{<n}^p; Y_{<n})$, and we set $\Delta_n := E_n(Y_{<n}^p; Y_{<n}) \bmod p$, a polynomial in $Y_0, \dots, Y_{n-1}$ only. It has no constant term because E_n has none (Lemma 2). $\square$

Lemma 5 (kernel of $\wp$). *If A is an integral domain over $\mathbb{F}_p$, then $\ker(\wp: W_r(A) \rightarrow W_r(A)) = W_r(\mathbb{F}_p) \cong \mathbb{Z}/p^r\mathbb{Z}$.*

Proof. $\wp(\underline{a}) = \underline{0}$ iff $F\underline{a} = \underline{a}$, i.e. $a_l^p = a_l$ for all l (both maps are componentwise), i.e. $a_l \in \mathbb{F}_p$ for all l since A is a domain. Finally $W_r(\mathbb{F}_p) = W(\mathbb{F}_p)/V^r W(\mathbb{F}_p) = \mathbb{Z}_p/p^r\mathbb{Z}_p \cong \mathbb{Z}/p^r\mathbb{Z}$ [Ser79, II §6]. $\square$

From now on we identify $G := \mathbb{Z}/p^r\mathbb{Z} = W_r(\mathbb{F}_p)$.

Proposition 6 (the Artin–Schreier–Witt isogeny is étale). *Let A be an $\mathbb{F}_p$ -algebra and $\underline{a} \in W_r(A)$. Set*

$$A_{\underline{a}} := A[Y_0, \dots, Y_{r-1}] / (\wp(\underline{Y})_n - a_n : 0 \leq n \leq r-1).$$

Then:

1. $A_{\underline{a}}$ is a free A -module of rank p^r , with basis the monomials $\prod_l Y_l^{c_l}$, $0 \leq c_l \leq p-1$;
2. $A_{\underline{a}}$ is étale over A ;
3. the rule $\underline{Y} \mapsto \underline{Y} +_W \underline{g}$, $\underline{g} \in G = W_r(\mathbb{F}_p)$, defines an action of G on $A_{\underline{a}}$ by A -algebra automorphisms, and if $A \neq 0$ this makes $\mathcal{P}_{\underline{a}} := \text{Spec } A_{\underline{a}}$ a G -torsor over $\text{Spec } A$ in the étale topology.

Proof. By Lemma 4 the defining equations are

$$Y_n^p - Y_n = a_n - \Delta_n(Y_0, \dots, Y_{n-1}), \quad n = 0, \dots, r-1,$$

a *triangular tower* of Artin–Schreier equations: each is monic of degree p in Y_n with coefficients in $A[Y_0, \dots, Y_{n-1}]$. (1) follows at once.

(2) $A_{\underline{a}}$ is flat (free) over A ; it remains to see it is unramified, i.e. $\Omega_{A_{\underline{a}}/A}^1 = 0$. The module $\Omega_{A_{\underline{a}}/A}^1$ is generated by $dY_0, \dots, dY_{r-1}$ subject to the relations $\sum_m (\partial g_n / \partial Y_m) dY_m = 0$, where $g_n = Y_n^p - Y_n + \Delta_n - a_n$. The Jacobian matrix $(\partial g_n / \partial Y_m)_{n,m}$ is lower triangular with diagonal entries $pY_n^{p-1} - 1 = -1$, hence invertible; so $\Omega_{A_{\underline{a}}/A}^1 = 0$.

(3) For $\underline{g} \in W_r(\mathbb{F}_p)$ we have $F\underline{g} = \underline{g}$, so $\wp(\underline{Y} +_W \underline{g}) = \wp(\underline{Y}) +_W \wp(\underline{g}) = \wp(\underline{Y})$; hence the substitution $\underline{Y} \mapsto \underline{Y} +_W \underline{g}$ preserves the defining ideal and gives an A -algebra endomorphism, with inverse given by $-\underline{g}$; and $(\underline{g}, \underline{g}') \mapsto \underline{g} +_W \underline{g}'$ makes this an action of G . For the torsor property it suffices, since $\mathcal{P}_{\underline{a}} \rightarrow \text{Spec } A$ is finite étale and surjective of degree $p^r = |G|$, to check that G acts simply transitively on geometric fibers. Let Ω be an algebraically closed A -field; the Ω -points of the fiber are the solutions $\underline{b} \in W_r(\Omega)$ of $\wp(\underline{b}) = \underline{a}$. If $\underline{b}, \underline{b}'$ are two solutions then $\wp(\underline{b}' -_W \underline{b}) = \underline{0}$, so $\underline{b}' -_W \underline{b} \in W_r(\mathbb{F}_p) = G$ by Lemma 5: the action is transitive; and it is free because translation by $\underline{g} \neq \underline{0}$ in the group $W_r(\Omega)$ has no fixed points. $\square$

Corollary 7 (integral datum $\Rightarrow$ unramified). *Let $\mathcal{O}$ be a discrete valuation ring containing $\mathbb{F}_p$, with fraction field K . If $\underline{a} \in W_r(\mathcal{O}) \subset W_r(K)$, then every field factor of the étale K -algebra $K_{\underline{a}} = K \otimes_{\mathcal{O}} \mathcal{O}_{\underline{a}}$ is an unramified extension of K with respect to $\mathcal{O}$ (in the sense of ??).*

Proof. By Proposition 6, $B := \mathcal{O}_{\underline{a}}$ is finite étale over $\mathcal{O}$. Being étale over a normal domain, B is a finite product $\prod_j B_j$ of normal domains, each finite étale over $\mathcal{O}$; then B_j is the integral closure of $\mathcal{O}$ in the field factor $L_j = \text{Frac}(B_j)$ of $K_{\underline{a}}$, it is a semi-local Dedekind domain, and at each of its maximal ideals the extension of discrete valuation rings is flat and unramified: the ramification index is 1 and the residue field extension is finite separable. This is precisely the definition of L_j/K being unramified with respect to $\mathcal{O}$. $\square$

2 Background II: Artin–Schreier–Witt theory

2.1 Classification of $\mathbb{Z}/p^r\mathbb{Z}$ -torsors on affine schemes

Proposition 8. *Let $U = \text{Spec } A$ be an integral affine scheme over $\mathbb{F}_p$. Then Proposition 6 induces a bijection*

$$W_r(A) / \wp W_r(A) \xrightarrow{\cong} \text{Tors}(U_{\text{ét}}, \mathbb{Z}/p^r\mathbb{Z}), \quad [\underline{a}] \mapsto [\mathcal{P}_{\underline{a}}].$$

Moreover the bijection is functorial: for a morphism $\text{Spec } B \rightarrow \text{Spec } A$ the pullback of $\mathcal{P}_{\underline{a}}$ is $\mathcal{P}_{\underline{a}'}$ where $\underline{a}'$ is the image of $\underline{a}$ in $W_r(B)$.

Proof. The sequence of abelian sheaves on $U_{\text{ét}}$

$$0 \longrightarrow \mathbb{Z}/p^r\mathbb{Z} \longrightarrow W_r(\mathcal{O}_U) \xrightarrow{\wp} W_r(\mathcal{O}_U) \longrightarrow 0$$

is exact: the kernel is the constant sheaf $\mathbb{Z}/p^r\mathbb{Z}$ by Lemma 5 (sections over a connected affine étale $V \rightarrow U$ with $V = \text{Spec } B$, B a domain — and every étale V is a disjoint union of such); and $\wp$ is surjective as a map of étale sheaves because for any $\underline{a} \in W_r(B)$ the finite étale surjective cover $\text{Spec } B_{\underline{a}} \rightarrow \text{Spec } B$ of Proposition 6 carries a tautological solution $\underline{Y}$ of $\wp(\underline{Y}) = \underline{a}$.

The long exact cohomology sequence gives

$$W_r(A) \xrightarrow{\varphi} W_r(A) \xrightarrow{\delta} H^1(U_{\text{ét}}, \mathbb{Z}/p^r\mathbb{Z}) \longrightarrow H^1(U_{\text{ét}}, W_r(\mathcal{O}_U)),$$

and the boundary map δ sends $\underline{a}$ to the class of the sheaf of solutions of $\wp(\underline{Y}) = \underline{a}$, i.e. of the torsor $\mathcal{P}_{\underline{a}}$. It remains to show $H^1(U_{\text{ét}}, W_r(\mathcal{O}_U)) = 0$. For $r = 1$, $W_1(\mathcal{O}_U) = \mathbb{G}_a$ and $H^1(U_{\text{ét}}, \mathbb{G}_a) \cong H^1(U, \mathcal{O}_U) = 0$ since U is affine (étale and Zariski cohomology of quasi-coherent sheaves agree, [Mil80, III §3]). For $r > 1$, Lemma 3(2) shows that $\{(0, \dots, 0, a)\} \subset W_r$ is a subgroup isomorphic to $\mathbb{G}_a$ and that truncation $(a_0, \dots, a_{r-1}) \mapsto (a_0, \dots, a_{r-2})$ is a group homomorphism (by Lemma 3(1), level- n components of a sum depend only on levels $\leq n$), giving an exact sequence of étale sheaves $0 \rightarrow \mathbb{G}_a \rightarrow W_r(\mathcal{O}_U) \rightarrow W_{r-1}(\mathcal{O}_U) \rightarrow 0$; conclude by induction. Functoriality of the construction $\underline{a} \mapsto A_{\underline{a}}$ is clear. $\square$

Lemma 9 (torsor invariants). *Let Γ be a finite group and $\text{Spec } B \rightarrow \text{Spec } A$ a Γ -torsor. Then $B^\Gamma = A$.*

Proof. $A \rightarrow B$ is finite étale, in particular faithfully flat, and the torsor condition gives a Γ -equivariant isomorphism of B -algebras $B \otimes_A B \cong \text{Maps}(\Gamma, B)$, $b \otimes b' \mapsto (\gamma \mapsto \gamma(b)b')$, where Γ acts on the left tensor factor and, on the right side, by permuting arguments (γ_0 sends a map φ to $\gamma \mapsto \varphi(\gamma\gamma_0)$). Taking Γ -invariants, which commutes with the flat base change $A \rightarrow B$,

$$B^\Gamma \otimes_A B = (B \otimes_A B)^{\Gamma \otimes 1} \cong \text{Maps}(\Gamma, B)^\Gamma = \{\text{constant maps}\} = B = A \otimes_A B.$$

By faithful flatness of B over A , $B^\Gamma = A$. $\square$

Lemma 10 (additivity: contracted products). *In the situation of Proposition 8, for $\underline{a}, \underline{b} \in W_r(A)$ there is a canonical isomorphism of G -torsors*

$$\mathcal{P}_{\underline{a}} \otimes^G \mathcal{P}_{\underline{b}} \cong \mathcal{P}_{\underline{a} +_W \underline{b}}.$$

Proof. Write the contracted product as $(\mathcal{P}_{\underline{a}} \times_U \mathcal{P}_{\underline{b}})/G^-$, where $G^- = \{(\underline{g}, -\underline{g})\} \leq G \times G$; the ring of $\mathcal{P}_{\underline{a}} \times_U \mathcal{P}_{\underline{b}}$ is

$$T = A[\underline{X}_1, \underline{X}_2] / (\wp(\underline{X}_1) - \underline{a}, \wp(\underline{X}_2) - \underline{b}),$$

where, here and below, “ $\wp(\underline{X}) - \underline{c}$ ” stands for the r componentwise equations, and $\underline{X}_1, \underline{X}_2$ are two batches of r variables.

Change of variables. Put $\underline{Y} := \underline{X}_1 +_W \underline{X}_2$. By Lemma 3(1), $Y_l = X_{1,l} + (\text{polynomial in } X_{1,<l} \text{ and } X_{2,\leq l})$, so the substitution $X_{1,l} \mapsto Y_l$ ($l = 0, \dots, r-1$) is a triangular, unipotent change of polynomial generators: $A[\underline{X}_1, \underline{X}_2] = A[\underline{Y}, \underline{X}_2]$.

The ideal in the new variables. We claim

$$(\wp(\underline{X}_1) - \underline{a}, \wp(\underline{X}_2) - \underline{b}) = (\wp(\underline{Y}) - (\underline{a} +_W \underline{b}), \wp(\underline{X}_2) - \underline{b}).$$

Both inclusions follow from the elementary fact that for any polynomial P and tuples u, v one has $P(u) - P(v) \in (u_1 - v_1, u_2 - v_2, \dots)$, combined with Witt-vector identities: for “ $\subseteq$ ” use $\underline{X}_1 = \underline{Y} -_W \underline{X}_2$ and the identity $\wp(\underline{Y} -_W \underline{X}_2) = \wp(\underline{Y}) -_W \wp(\underline{X}_2)$ (as polynomial identities in the components — $\wp$ is additive), evaluating $P =$ each component of $\wp(\cdot -_W \cdot)$ at $u = (\wp \underline{Y}, \wp \underline{X}_2)$ and $v = (\underline{a} +_W \underline{b}, \underline{b})$, and noting $(\underline{a} +_W \underline{b}) -_W \underline{b} = \underline{a}$; the inclusion “ $\supseteq$ ” is the same computation run backwards, with $P =$ components of $\wp(\cdot +_W \cdot)$. Hence

$$T \cong A_{\underline{a} +_W \underline{b}} \otimes_A A_{\underline{b}}, \quad \underline{Y} \mapsto \underline{Y} \otimes 1, \quad \underline{X}_2 \mapsto 1 \otimes \underline{X}_2.$$

Invariants. The subgroup G^- acts by $\underline{Y} \mapsto \underline{Y} +_W \underline{g} -_W \underline{g} = \underline{Y}$ and $\underline{X}_2 \mapsto \underline{X}_2 -_W \underline{g}$; i.e. trivially on the first tensor factor and through the full torsor action of G on the second. By Lemma 9, $T^{G^-} = A_{\underline{a}+_W \underline{b}} \otimes_A (A_{\underline{b}})^G = A_{\underline{a}+_W \underline{b}}$. The residual $G = (G \times G)/G^-$ -action is $\underline{Y} \mapsto \underline{Y} +_W \underline{g}$, as required. $\square$

2.2 Local theory over $k((x))$: reduced data and ramification breaks

Let k be an algebraically closed field of characteristic p , let $K = k((x))$ with valuation v , valuation ring $k[[x]]$ and maximal ideal $\mathfrak{m}_{k[[x]]} = xk[[x]]$. By Proposition 8 applied to Galois cohomology (or see [Tho05]), continuous characters $\rho: \text{Gal}(K^{sep}/K) \rightarrow \mathbb{Z}/p^r\mathbb{Z}$ correspond to classes $[\underline{f}] \in W_r(K)/\wp W_r(K)$; the fixed field of $\ker \rho$ is the splitting field of $\wp(\underline{Y}) = \underline{f}$.

Definition 11 (reduced datum). A Witt vector $\underline{f} = (f_0, \dots, f_{r-1}) \in W_r(K)$ is *reduced* if every component f_l is a polynomial in x^{-1} without constant term, and its degree $m_l := \deg_{x^{-1}} f_l$ (the pole order of f_l ; set $m_l = 0$ if $f_l = 0$) is either 0 or prime to p . We call

$$\mu(\underline{f}) := \max_{0 \leq l \leq r-1} p^{r-1-l} m_l$$

the *weighted pole order* of $\underline{f}$.

Lemma 12 (reduction to reduced form). *Every class in $W_r(K)/\wp W_r(K)$ has a reduced representative.*

Proof. We first record two solvability facts, both immediate from the triangular shape of $\wp$ (Lemma 4).

(a) $\wp: W_r(\mathfrak{m}_{k[[x]]}) \rightarrow W_r(\mathfrak{m}_{k[[x]]})$ is surjective. Note $W_r(\mathfrak{m}_{k[[x]]}) := \{\underline{c} : c_l \in \mathfrak{m}_{k[[x]]} \ \forall l\}$ is a subgroup of $W_r(k[[x]])$, since the addition polynomials have no constant term. Given $\underline{c} \in W_r(\mathfrak{m}_{k[[x]]})$, solve $b_n^p - b_n = c_n - \Delta_n(b_{<n})$ recursively: assuming $b_0, \dots, b_{n-1} \in \mathfrak{m}_{k[[x]]}$, the right-hand side $t := c_n - \Delta_n(b_{<n})$ lies in $\mathfrak{m}_{k[[x]]}$ (Δ_n has no constant term), and $b_n := -(t + t^p + t^{p^2} + \dots) \in \mathfrak{m}_{k[[x]]}$ converges x -adically and satisfies $b_n^p - b_n = t$.

(b) $\wp: W_r(k) \rightarrow W_r(k)$ is surjective, since k is algebraically closed and each equation $b_n^p - b_n = c_n - \Delta_n(b_{<n})$ is solvable in k .

Now let $\underline{a} \in W_r(K)$. We normalize the components from level 0 upwards; the key point, guaranteed by Lemma 3(2), is that subtracting $\wp(V^n(h))$, where $V^n(h) := (0, \dots, 0, h, 0, \dots, 0)$ (h in level n), changes the level- n component by exactly $-\wp(h) = -(h^p - h)$, changes *no* component of level $< n$, and changes the components of level $> n$ by elements of K .

So suppose levels $0, \dots, n-1$ are already reduced. Write $a_n = a_n^- + c + a_n^+$ with $a_n^- \in x^{-1}k[x^{-1}]$, $c \in k$, $a_n^+ \in \mathfrak{m}_{k[[x]]}$.

- By (a) and (b) choose h', h'' with $\wp(h') = a_n^+$ ($h' \in \mathfrak{m}_{k[[x]]}$) and $\wp(h'') = c$ ($h'' \in k$); subtracting $\wp(V^n(h' + h''))$ we may assume $a_n = a_n^-$.
- If the pole order m of a_n is positive and divisible by p , with leading term cx^{-m} , subtract $\wp(V^n(c^{1/p}x^{-m/p}))$: the level- n component changes by $-(cx^{-m} - c^{1/p}x^{-m/p})$, which strictly decreases its pole order. Iterate; after finitely many steps, and after removing once more the constant and positive parts as above, the level- n component is a polynomial in x^{-1} without constant term, of degree 0 or prime to p .

All modifications leave levels $< n$ untouched, so the process terminates after treating level $r-1$. $\square$

The following classical theorem of Schmid and Witt describes the ramification of the corresponding extensions; it is the promised Artin–Schreier–Witt input. For $r = 1$ it is exactly ??.

Theorem 13 (Schmid–Witt; [Tho05]). *Let $\underline{f} \in W_r(K)$ be reduced with weighted pole order $\mu = \mu(\underline{f})$, and let $\rho: \text{Gal}(K^{\text{sep}}/K) \rightarrow \mathbb{Z}/p^r\mathbb{Z}$ be the character of the class $[\underline{f}]$. Then, in the upper-numbering convention of ??:*

1. *if $\mu = 0$ then $\rho(\text{Gal}(K^{\text{sep}}/K)^{(u)}) = 1$ for all $u \geq 1$ (the extension cut out by ρ is unramified);*
2. *if $\mu > 0$ then the largest upper ramification break of the extension cut out by ρ equals μ ; that is, $\rho(\text{Gal}(K^{\text{sep}}/K)^{(\mu)}) \neq 1$ and $\rho(\text{Gal}(K^{\text{sep}}/K)^{(u)}) = 1$ for $u > \mu$.*

In particular, in the sense of ??, the class $[\underline{f}]$ has ramification bounded by d if and only if $\mu(\underline{f}) \leq d-1$.

Remark. Only the implication “ramification bounded by $d \Rightarrow \mu \leq d-1$ ” (i.e. the sharpness statement in (2)) is used below; the easy converse direction is not needed. Note also that Theorem 13 does not require ρ to be surjective: if the extension cut out by ρ has degree $p^{r'} < p^r$, the statement is consistent with itself under the shift identification $W_{r'} \hookrightarrow W_r$ (Verschiebung), since μ is invariant under it.

3 The Artin–Schreier–Witt datum of the symmetric power

3.1 Setting and reductions

We place ourselves in the situation of ??, whose notation we keep: $G = \mathbb{Z}/p^r\mathbb{Z}$; $\mathcal{P} \rightarrow U = C \setminus \mathfrak{m}$ has ramification bounded by $\mathfrak{m}$; $P \in \text{Supp } \mathfrak{m}$ with multiplicity n ; $\mathfrak{n} = nP$ and $d := \deg \mathfrak{n} = n$; and $\eta_{\mathfrak{n}}$ is the generic point of the exceptional divisor of the blowup of $C^{(d)}$ at $\mathfrak{n}$.

As in the thesis we may assume k algebraically closed (étale base change to $k^{\text{sep}} = \bar{k}$ does not change any of the ramification filtrations involved). By the principle of invariance of local invariants under formal isomorphism (??, discussion preceding the $r = 1$ theorem) the ramification of $\mathcal{P}^{[d]}$ at $\eta_{\mathfrak{n}}$ depends only on the completed local ring $\hat{\mathcal{O}}_{C,P} \cong k[[x]]$ together with the restriction of $\mathcal{P}$ to its punctured formal neighborhood, i.e. on the class

$$[\underline{a}] \in W_r(k((x)))/\wp W_r(k((x)))$$

corresponding to $\mathcal{P}|_{\text{Spec } \hat{K}_P}$ under Proposition 8 (equivalently, on the local character ρ of $G_{\hat{K}_P}$). By Lemma 12 choose a *reduced* representative $\underline{f}$ of $[\underline{a}]$; since $\mathcal{P}$ has ramification bounded by d at P , Theorem 13 gives

$$\mu(\underline{f}) = \max_{0 \leq l \leq r-1} p^{r-1-l} m_l \leq d-1. \quad (3)$$

Now set $y := x^{-1}$ and consider

$$C = \mathbb{P}_k^1, \quad P = 0, \quad U = \mathbb{P}_k^1 \setminus \{0\} = \text{Spec } k[y] \cong \mathbb{A}_k^1.$$

The components f_l of $\underline{f}$ are polynomials in y without constant term, so $\underline{f} \in W_r(k[y])$ and $\mathcal{Q} := \mathcal{P}_{\underline{f}} \rightarrow U$ (Proposition 6) is a G -torsor on U whose completed local datum at 0 is $[\underline{f}] = [\underline{a}]$. By the invariance principle it therefore suffices to prove the theorem for $(\mathbb{P}^1, d \cdot 0, \mathcal{Q})$; we assume from now on that

$$\mathcal{P} = \mathcal{P}_{\underline{f}} \text{ over } U = \text{Spec } k[y], \quad \underline{f} \text{ reduced}, \quad \mu(\underline{f}) \leq d-1.$$

Remark (the case $C = \mathbb{P}^1$ needs no formal-invariance principle). When $C = \mathbb{P}^1$ to begin with, the replacement of $\mathcal{P}$ by $\mathcal{P}_f$ is an honest isomorphism of torsors and the invariance principle is not needed. Indeed the natural map

$$W_r(k[y])/\wp W_r(k[y]) \longrightarrow W_r(k((x)))/\wp W_r(k((x)))$$

is *injective*: suppose $\underline{g} \in W_r(k[y])$ and $\underline{g} = \wp(\underline{h})$ with $\underline{h} \in W_r(k((x)))$. At level 0, $h_0^p - h_0 = g_0 \in k[y] = k[x^{-1}]$; writing $h_0 = h^- + c + h^+$ (pole part, constant, part in $\mathfrak{m}_{k[[x]]}$) and using $\wp(h_0) = \wp(h^-) + \wp(c) + \wp(h^+)$ (Frobenius is additive), the $\mathfrak{m}_{k[[x]]}$ -part of g_0 is $\wp(h^+) = 0$; if $h^+ \neq 0$ its lowest-degree term survives in $h^{+p} - h^+$, so $h^+ = 0$ and $h_0 \in k[y]$. Inductively, at level n , $h_n^p - h_n = g_n - \Delta_n(h_{<n}) \in k[y]$ (Lemma 4, with $h_{<n} \in k[y]$ by induction), and the same argument gives $h_n \in k[y]$. Hence $\underline{g} \in \wp W_r(k[y])$. Consequently any two Witt vectors in $W_r(k[y])$ with the same class over $k((x))$ define isomorphic torsors on U (Proposition 8).

3.2 Coordinates on the symmetric power and the blowup

Since $U = \text{Spec } k[y] \cong \mathbb{A}^1$, we have

$$U^d = \text{Spec } k[y_1, \dots, y_d], \quad U^{(d)} = \text{Spec } k[\varepsilon_1, \dots, \varepsilon_d], \quad \varepsilon_j := e_j(y_1, \dots, y_d),$$

with e_j the elementary symmetric polynomials. On the other side, x is a coordinate at $P = 0$, the point $\mathfrak{n} = d \cdot 0 \in C^{(d)}$ has the affine neighborhood $(\mathbb{A}_x^1)^{(d)} = \text{Spec } k[e_1, \dots, e_d]$ with $e_j := e_j(x_1, \dots, x_d)$ vanishing at $\mathfrak{n}$, and (as computed in ?? for the blowup of affine space at the origin) the local ring $R := \mathcal{O}_{X_{\mathfrak{n}}, \eta_{\mathfrak{n}}}$ is a discrete valuation ring with

$$v_{\eta}(e_j) = 1 \ (1 \leq j \leq d), \quad \kappa(\eta_{\mathfrak{n}}) = k\left(\frac{e_1}{e_d}, \dots, \frac{e_{d-1}}{e_d}\right), \quad \widehat{R} = \kappa(\eta_{\mathfrak{n}})[[e_d]].$$

Since $y_i = 1/x_i$, the two coordinate systems are related by

$$\varepsilon_j = e_j(y_1, \dots, y_d) = \frac{e_{d-j}(x_1, \dots, x_d)}{e_d(x_1, \dots, x_d)} = \frac{e_{d-j}}{e_d} \quad (1 \leq j \leq d, \ e_0 := 1). \quad (4)$$

In particular:

$$\varepsilon_j \in \kappa(\eta_{\mathfrak{n}}) \subset \widehat{R} \quad \text{for } 1 \leq j \leq d-1, \quad \text{while} \quad \varepsilon_d = \frac{1}{e_d} \text{ has } v_{\eta}(\varepsilon_d) = -1. \quad (5)$$

Both systems generate the same function field $K(C^{(d)}) = k(e_1, \dots, e_d) = k(\varepsilon_1, \dots, \varepsilon_d)$, and we write $\widehat{R}_{\eta} = \kappa(\eta_{\mathfrak{n}})((e_d)) = \text{Frac}(\widehat{R})$.

3.3 The datum of $\mathcal{P}^{[d]}$ is the d -fold Witt sum

Recall (??) that $\mathcal{P}^{[d]} = \mathcal{P}^{\times d}/\Xi$ with $\Xi = K_G \rtimes S_d$, where $K_G = \ker(\Sigma: G^d \rightarrow G)$; for affine U this quotient is the spectrum of the ring of invariants.

Theorem 14 (the symmetric-power datum). *With $\mathcal{P} = \mathcal{P}_f$ over $U = \text{Spec } k[y]$ as in Section 3.1, there is an isomorphism of G -torsors over $U^{(d)} = \text{Spec } k[\varepsilon_1, \dots, \varepsilon_d]$*

$$\mathcal{P}^{[d]} \cong \mathcal{P}_{\underline{F}}, \quad \text{where} \quad \underline{F} := \sum_{i=1}^d \underline{f}(y_i) \in W_r(k[\varepsilon_1, \dots, \varepsilon_d])$$

is the d -fold Witt sum of the pullbacks of $\underline{f}$ along the d projections. (Its components $F_j = S_j^{(d)}(\underline{f}(y_1), \dots, \underline{f}(y_d))$ are symmetric polynomials in $y_1, \dots, y_d$, hence indeed lie in $k[\varepsilon_1, \dots, \varepsilon_d]$.)

Proof. Write $A^{\otimes d} = k[y_1, \dots, y_d]$. The fiber product $\mathcal{P}^{\times d} = p_1^{-1}\mathcal{P} \times_{U^d} \dots \times_{U^d} p_d^{-1}\mathcal{P}$ is the spectrum of

$$T = A^{\otimes d}[\underline{X}_1, \dots, \underline{X}_d] / (\wp(\underline{X}_i) - \underline{f}(y_i) : 1 \leq i \leq d),$$

a G^d -torsor over U^d , where $\underline{g}_i \in G$ acts by $\underline{X}_i \mapsto \underline{X}_i +_W \underline{g}_i$, and S_d acts by permuting simultaneously the y_i and the batches $\underline{X}_i$.

Step 1: K_G -invariants. Put $\underline{Y} := \sum_{i=1}^{W^d} \underline{X}_i$. Exactly as in the proof of [Lemma 10](#) (using [Lemma 3](#)(1) for the batch $\underline{X}_1$), the substitution $\underline{X}_{1,l} \mapsto Y_l$ is a triangular unipotent change of polynomial generators, $A^{\otimes d}[\underline{X}_1, \dots, \underline{X}_d] = A^{\otimes d}[\underline{Y}, \underline{X}_2, \dots, \underline{X}_d]$, and the same two-sided ideal computation (with $\underline{X}_1 = \underline{Y} -_W \underline{X}_2 -_W \dots -_W \underline{X}_d$ and additivity of $\wp$) identifies the defining ideal with

$$(\wp(\underline{Y}) - \underline{F}, \wp(\underline{X}_j) - \underline{f}(y_j) : 2 \leq j \leq d).$$

Hence

$$T \cong \underbrace{A^{\otimes d}[\underline{Y}] / (\wp(\underline{Y}) - \underline{F})}_{=: B} \otimes_{A^{\otimes d}} T', \quad T' := A^{\otimes d}[\underline{X}_2, \dots, \underline{X}_d] / (\wp(\underline{X}_j) - \underline{f}(y_j)),$$

and $\text{Spec } T' \rightarrow U^d$ is a G^{d-1} -torsor ([Proposition 6](#)). Parametrize $K_G \cong G^{d-1}$ by $(\underline{g}_2, \dots, \underline{g}_d) \mapsto (-\sum_{j \geq 2}^W \underline{g}_j, \underline{g}_2, \dots, \underline{g}_d)$. Under this action $\underline{Y} \mapsto \underline{Y} +_W \sum_i^W \underline{g}_i = \underline{Y}$ is invariant, while $(\underline{g}_2, \dots, \underline{g}_d)$ acts on T' through the full torsor action of G^{d-1} . By [Lemma 9](#),

$$T^{K_G} = B \otimes_{A^{\otimes d}} (T')^{G^{d-1}} = B \otimes_{A^{\otimes d}} A^{\otimes d} = B.$$

The residual action of $G = G^d/K_G$ on B is $\underline{Y} \mapsto \underline{Y} +_W \underline{g}$.

Step 2: S_d -invariants. The S_d -action on T preserves the subring $B = T^{K_G}$ (it normalizes K_G in Ξ); explicitly it permutes the y_i and fixes each component Y_l of $\underline{Y}$ (the Witt sum $\sum_i^W \underline{X}_i$ is invariant under permuting its summands), and it fixes each component of $\underline{F}$ (a symmetric polynomial). By [Proposition 6](#)(1), B is a free $A^{\otimes d}$ -module with basis the monomials $\prod_l Y_l^{c_l}$ ($0 \leq c_l \leq p-1$), and S_d acts on B coefficientwise with respect to this basis. Hence

$$B^{S_d} = (A^{\otimes d})^{S_d}[\underline{Y}] / (\wp(\underline{Y}) - \underline{F}) = k[\varepsilon_1, \dots, \varepsilon_d][\underline{Y}] / (\wp(\underline{Y}) - \underline{F}),$$

which is precisely the ring of $\mathcal{P}_{\underline{F}}$ over $U^{(d)} = \text{Spec } k[\varepsilon_1, \dots, \varepsilon_d]$. Therefore $\mathcal{P}^{[d]} = \text{Spec } T^{\Xi} = \text{Spec } B^{S_d} = \mathcal{P}_{\underline{F}}$, compatibly with the G -actions. $\square$

Remark. For $r = 1$ this is literally the computation of the thesis (??): $\underline{Y}$ is the single variable $Y = X_1 + \dots + X_d$ and $\underline{F}$ the power sum $\sum_i f(y_i)$. The Witt sum is the correct p^r -analogue: the “carries” of Witt addition, which obstructed the inductive approach, are now explicit polynomial terms that the next two lemmas control.

4 Integrality at the exceptional divisor, and the proof

Lemma 15 (isobaric degree bound). *Let $\underline{f}$ be reduced with pole orders m_l and weighted pole order $\mu = \mu(\underline{f})$, and let $F_j = S_j^{(d)}(\underline{f}(y_1), \dots, \underline{f}(y_d))$ be the components of the Witt sum $\underline{F}$ of [Theorem 14](#). Then F_j is a symmetric polynomial in $y_1, \dots, y_d$ of total degree*

$$\deg F_j \leq p^{j-r+1} \mu \leq \mu \quad (0 \leq j \leq r-1).$$

Proof. By Lemma 2, every monomial of $S_j^{(d)}$ has the form $\prod_{i,l} (X_l^{(i)})^{n_{i,l}}$ with $\sum_{i,l} n_{i,l} p^l = p^j$. Substituting $X_l^{(i)} \mapsto f_l(y_i)$, a polynomial of degree m_l in y_i , the resulting term has total degree

$$\sum_{i,l} n_{i,l} m_l \leq \sum_{i,l} n_{i,l} p^l \cdot \max_l \frac{m_l}{p^l} = p^j \cdot p^{1-r} \max_l p^{r-1-l} m_l = p^{j-r+1} \mu \leq \mu,$$

using $j \leq r-1$. Symmetry of F_j is clear since permuting the y_i permutes the summands of the Witt sum. $\square$

Lemma 16 (low-degree symmetric polynomials are residue-field constants). *Let $F \in k[y_1, \dots, y_d]^{S_d}$ be a symmetric polynomial of total degree $D < d$. Then $F \in k[\varepsilon_1, \dots, \varepsilon_{d-1}]$ (ε_d does not occur), and consequently, under (4) and (5), the image of F in $\widehat{K}_\eta$ lies in the residue field $\kappa(\eta_{\mathfrak{n}}) \subset \widehat{R}$; in particular $v_\eta(F) \geq 0$.*

Proof. By the fundamental theorem of symmetric polynomials, F is a polynomial in $\varepsilon_1, \dots, \varepsilon_d$; we claim only $\varepsilon_1, \dots, \varepsilon_D$ occur. Recall the standard algorithm: if $y^\lambda = y_1^{\lambda_1} \cdots y_s^{\lambda_s}$ ($\lambda_1 \geq \dots \geq \lambda_s > 0$) is the lexicographically leading monomial of F , with coefficient c , one subtracts $c \varepsilon_1^{\lambda_1 - \lambda_2} \varepsilon_2^{\lambda_2 - \lambda_3} \cdots \varepsilon_s^{\lambda_s}$, obtaining a symmetric polynomial of the same or smaller degree with a lexicographically smaller leading term, and iterates. A monomial of degree $\leq D$ involves at most D distinct variables, so $s \leq D$ at every step, and only $\varepsilon_1, \dots, \varepsilon_D$ with $D < d$ are ever used. The last statement now follows from (5): each ε_j with $j \leq d-1$ maps to the element e_{d-j}/e_d of the coefficient field $\kappa(\eta_{\mathfrak{n}})$ of $\widehat{R} = \kappa(\eta_{\mathfrak{n}})[[e_d]]$. $\square$

Remark. The hypothesis $D < d$ is exactly what excludes $\varepsilon_d = 1/e_d$, the unique pole at $\eta_{\mathfrak{n}}$ among the ε_j ; this is the multi-level generalization of the thesis's $r = 1$ lemma on $\sum_i f(y_i)$, and it is sharp — see the concluding remarks.

Proof of ??

Proof. By the reductions of Section 3.1 we may assume $k = \bar{k}$, $C = \mathbb{P}^1$, $P = 0$, $U = \operatorname{Spec} k[y]$ and $\mathcal{P} = \mathcal{P}_{\underline{f}}$ with $\underline{f}$ reduced and $\mu(\underline{f}) \leq d-1$ (this used Lemma 12 and the Schmid–Witt bound, Theorem 13).

By Theorem 14, $\mathcal{P}^{[d]} \cong \mathcal{P}_{\underline{F}}$ over $U^{(d)}$ with $\underline{F} = \sum_{i=1}^{W^d} \underline{f}(y_i)$. Restricting to the completed field at the generic point of the exceptional divisor (Section 3.2), the torsor $\mathcal{P}^{[d]}|_{\operatorname{Spec} \widehat{K}_\eta}$ is $\mathcal{P}_{\underline{F}'}$ where $\underline{F}' \in W_r(\widehat{K}_\eta)$ is the image of $\underline{F}$.

By Lemma 15, each component F_j is a symmetric polynomial in $y_1, \dots, y_d$ of total degree $\leq \mu(\underline{f}) \leq d-1 < d$; hence by Lemma 16 each F'_j lies in $\kappa(\eta_{\mathfrak{n}}) \subset \widehat{R}$. Therefore

$$\underline{F}' \in W_r(\widehat{R}),$$

and by Corollary 7 every field factor of $\mathcal{P}_{\underline{F}'}$ is unramified over $\widehat{R}$. Since ramification at $\eta_{\mathfrak{n}}$ is computed exactly on this restriction (?? and the surrounding discussion), $\mathcal{P}^{[d]}$ is unramified at $\eta_{\mathfrak{n}}$. $\square$

Concluding remarks

Remark (sharpness, and the boundary examples). [Lemma 15](#) is in fact an equality, $\deg F_{r-1} = \mu(\underline{f})$: the extremal monomial $(X_{l_0}^{(i)})^{p^{r-1-l_0}}$ realizing $\mu = p^{r-1-l_0}m_{l_0}$ occurs in $S_{r-1}^{(d)}$ with a nonzero coefficient mod p , and its top-degree part cannot cancel. Hence the hypothesis $\mu \leq d-1$ is optimal: if $\mu \geq d$ the top component of the datum genuinely involves $\varepsilon_d = 1/e_d$ and acquires a pole at η_n . This matches the boundary phenomena recorded in the thesis: for $p = 2$, $r = 2$, $m_0 = 1$ (so $\mu = 2$) the symmetric square ($d = 2$) is ramified at η_n (??), while $d = 3 > \mu$ is unramified; and the machine verifications in `q_ramification_along_VF/general_witt_proof` (`witt_general.py`, `witt_r3.py`) confirm $\deg F_j = p^{j-r+1}\mu$ exactly for $r = 2, 3$ and various p .

Remark (why the induction stalled). The inductive route factored $\mathcal{P} \rightarrow \mathcal{P}' \rightarrow U$ through the intermediate curve and tried to reconstruct the ramification of $\mathcal{P}^{[d]}$ from that of $\mathcal{P}'^{[d]}$ across the base change q ; the loss of information there (the “absorption” phenomenon of `a_wrong_path.tex`) is precisely the Witt carry. In the present proof the carry is not estimated across a base change but computed in closed form inside the symmetric power, where the isobaric grading shows it obeys the same degree bound as the leading terms. No induction on r , and no Swan conductor theory, is needed.

- Remark* (integration into the thesis). 1. [Section 1](#) and [Section 2](#) belong in the Preliminaries, next to Kummer and Artin–Schreier theories (??, ??); [Theorem 13](#) generalizes the latter and can replace it.
2. [Sections 3.1](#) and [3.2](#), [Theorem 14](#), [Lemmas 15](#) and [16](#) and the final proof replace the unfinished proof of ?? in `RamificationAfterBlowupIsTame.tex`. Note that [Theorem 14](#) at $r = 1$ also subsumes the Artin–Schreier half of the theorem preceding ??, so the two proofs can be merged if desired.
3. The deduction of the general abelian case (??) is unchanged: decompose G by the structure theorem and combine the cyclic cases via ?? and ?? (or ??), exactly as outlined in the thesis.

References

- [Ser79] Jean-Pierre Serre. *Local Fields*. Springer, 1979.
- [Mil80] James S. Milne. *Étale Cohomology*. Vol. 33. Princeton Mathematical Series. Princeton, NJ: Princeton University Press, 1980.
- [Tho05] Lara Thomas. “Ramification groups in Artin-Schreier-Witt extensions”. In: *Journal de théorie des nombres de Bordeaux* 17.2 (2005), pp. 689–720. DOI: [10.5802/jtnb.514](https://doi.org/10.5802/jtnb.514). URL: <https://jtnb.centre-mersenne.org/articles/10.5802/jtnb.514/>.
- [Rab14] Joseph Rabinoff. *The Theory of Witt Vectors*. 2014. arXiv: [1409.7445](https://arxiv.org/abs/1409.7445) [[math.RA](#)]. URL: <https://arxiv.org/abs/1409.7445>.